Food and Agriculture Organization of the United Nations

Crop Module 2
AquaCrop input requirements

Jorge Alvar-Beltrán

(2023)

Content

Day 1 (introduction to AquaCrop)

- How do crop models work?
- AquaCrop interface
- AquaCrop: climate, crop, management and soil modules
- Run simulations and interpret results

Days 2 (Running your own simulation-standard mode)

- Input requirements (details)
- Create/import climatic files on AquaCrop

AquaCrop input requirements: climatic information

Input variables	Yes/No	Frequency
Precipitation	Yes	Daily, dekadal, monthly
Temperature (max/min)	Yes	Daily, dekadal, monthly
Reference evapotranspiration	Yes/pre-computed	Daily, dekadal, monthly
Relative humidity	+	Daily, dekadal, monthly
Solar radiation	+	Daily, dekadal, monthly
Wind speed	+	Daily, dekadal, monthly
CO_2 concentrations (RCPs 4.5, 8.5 etc.)	Available	Daily, dekadal, monthly

AquaCrop input requirements: crop information

Input variables	Yes/No	Frequency
Planting date	Yes	Once
Plant density	Yes	Once
Planting method	Yes	Once
Maximum canopy cover	Yes	Once
Time to emergence	Yes	Once
Time to & duration flowering	Yes	Once
Start of canopy senescence	Yes	Once
Crop coefficient	Yes	Once

AquaCrop input requirements: crop information

Input variables	Yes/No	Frequency
Time to maturity	Yes	Once
Length building up harvest index	+	Once
Crop water productivity	+	Once
Harvest index	Yes	Once
Root depth	+	Once
Time for maximum root depth	+	Once

AquaCrop input requirements: soil information

Input variables	Yes/No	Frequency
Soil texture	Yes/Available	Once
Soil water content (SAT, FC, PWP)	Available	Once
Effective soil depth	+	Once
Soil coarse material	+	Once
Drainage	+	Once
Electrical conductivity	+	Once
Groundwater table (depth)	+	Once
Groundwater table (salinity)	+	Once

AquaCrop input requirements: management info.

Input variables	Yes/No	Frequency
Soil fertility	+	Once
Cover & type of soil mulches	+	Once
Height of soil bunds	+	Once
Surface runoff	+	Once
Irrigation method	Yes (if irrigated)	Once
Application depth	Yes (if irrigated)	At each event
Time of irrigation	Yes (if irrigated)	At each event
Salinity of the irrigation	+	Once

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Crop Module 2
Creating Climate files

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(2023)

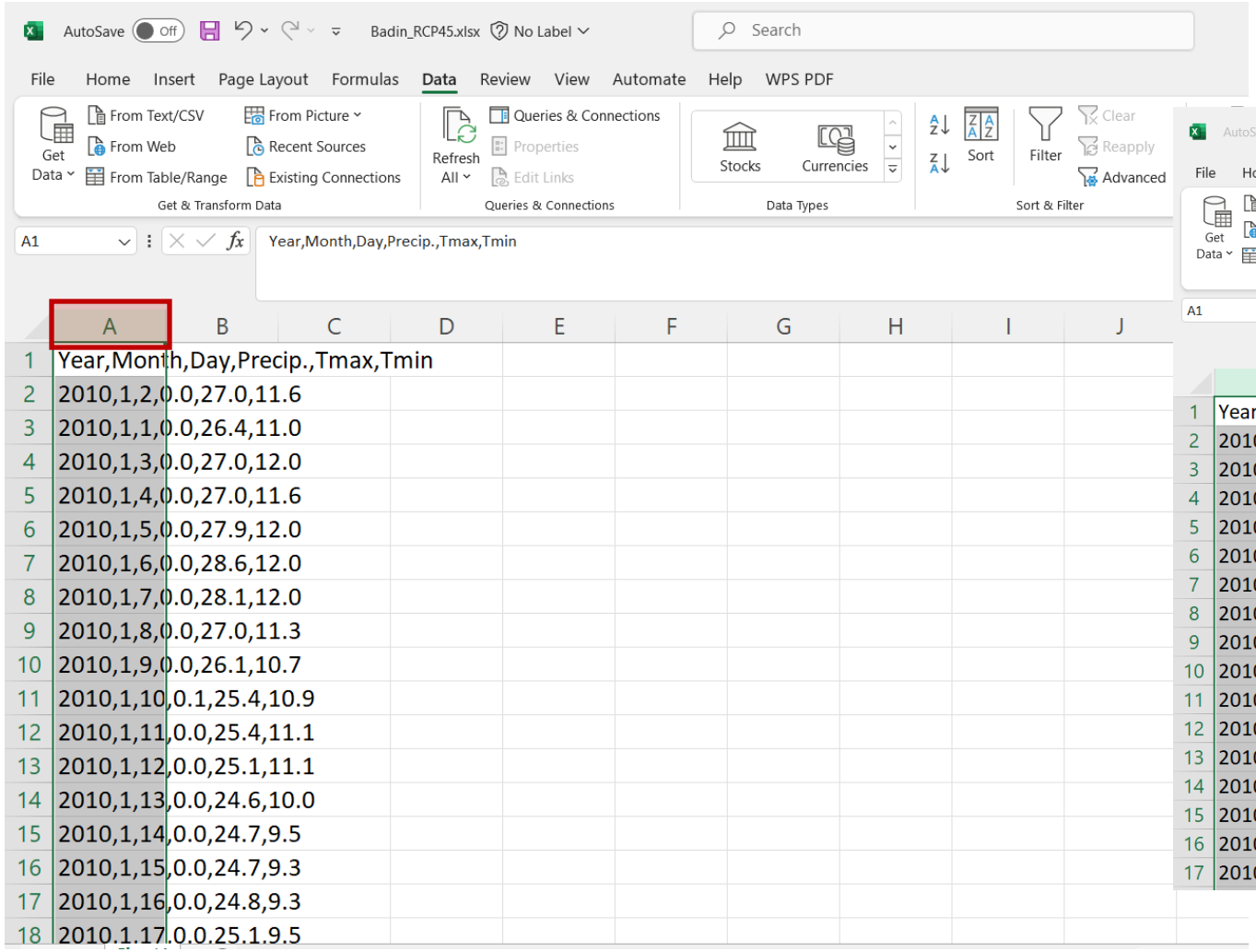
Data used in this session

Climate module: create our own climatic files (daily values) for Badin from 01-01-2010 until 31-12-2099.

- One climate model for RCP4.5, bias-corrected with W5E5
- Climatic variables: maximum temperature (Tmax), minimum temperature (Tmin), precipitation (pr).

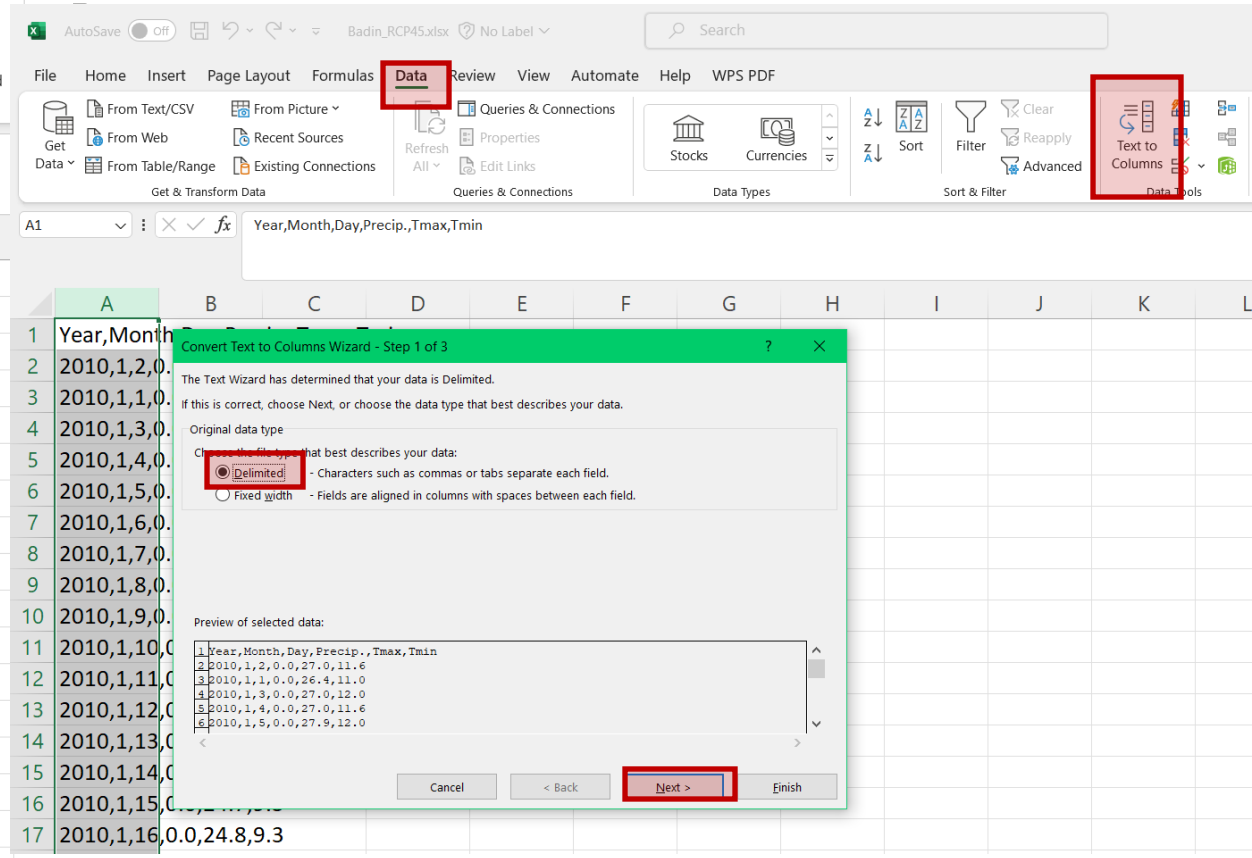
Preparing the CSV climatic files

Step 1: select column **A**



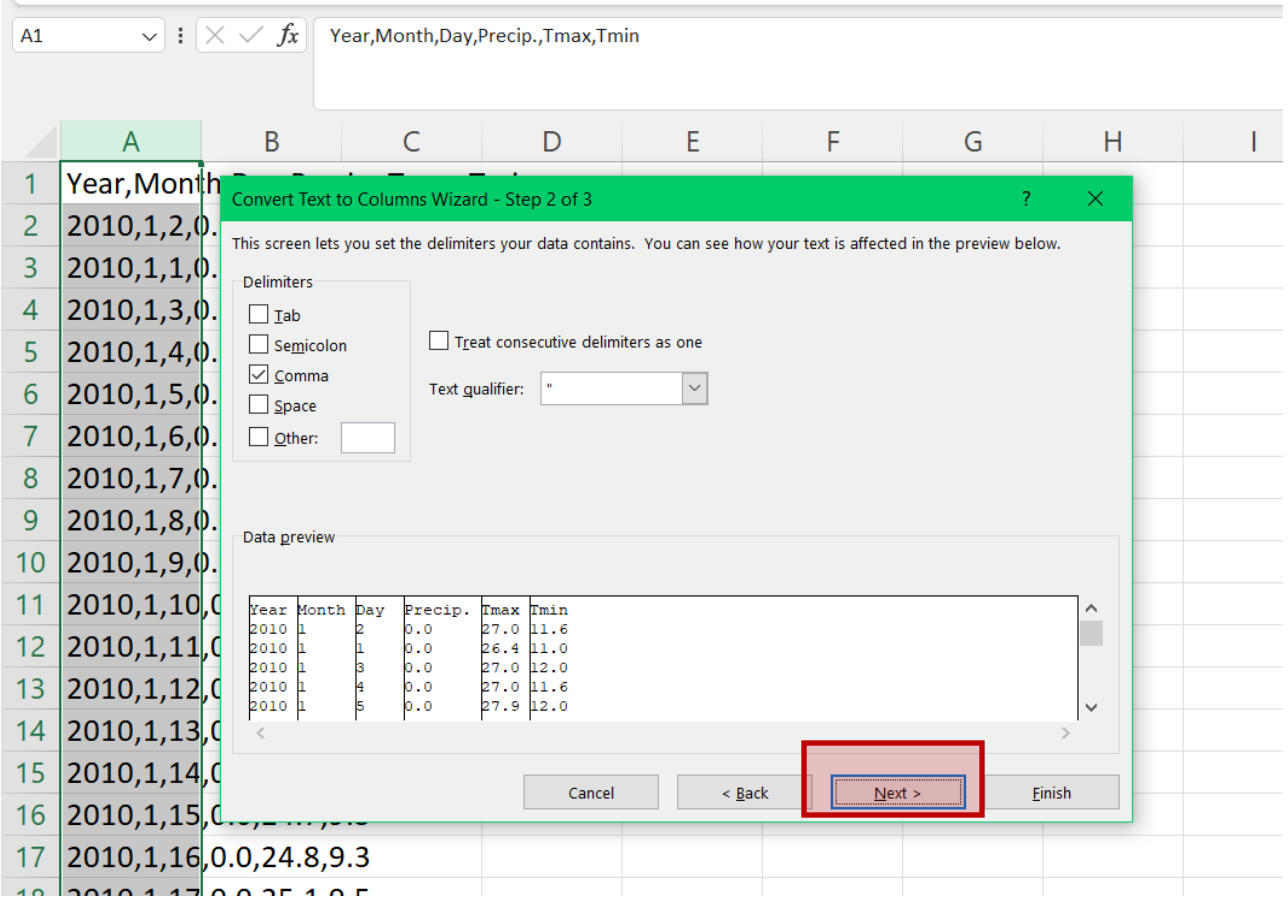
Step 2: click the **Data tab** in the upper toolbar and, afterwards, click in **Text to Columns**

Step 3: click on **Delimited** and then **Next**

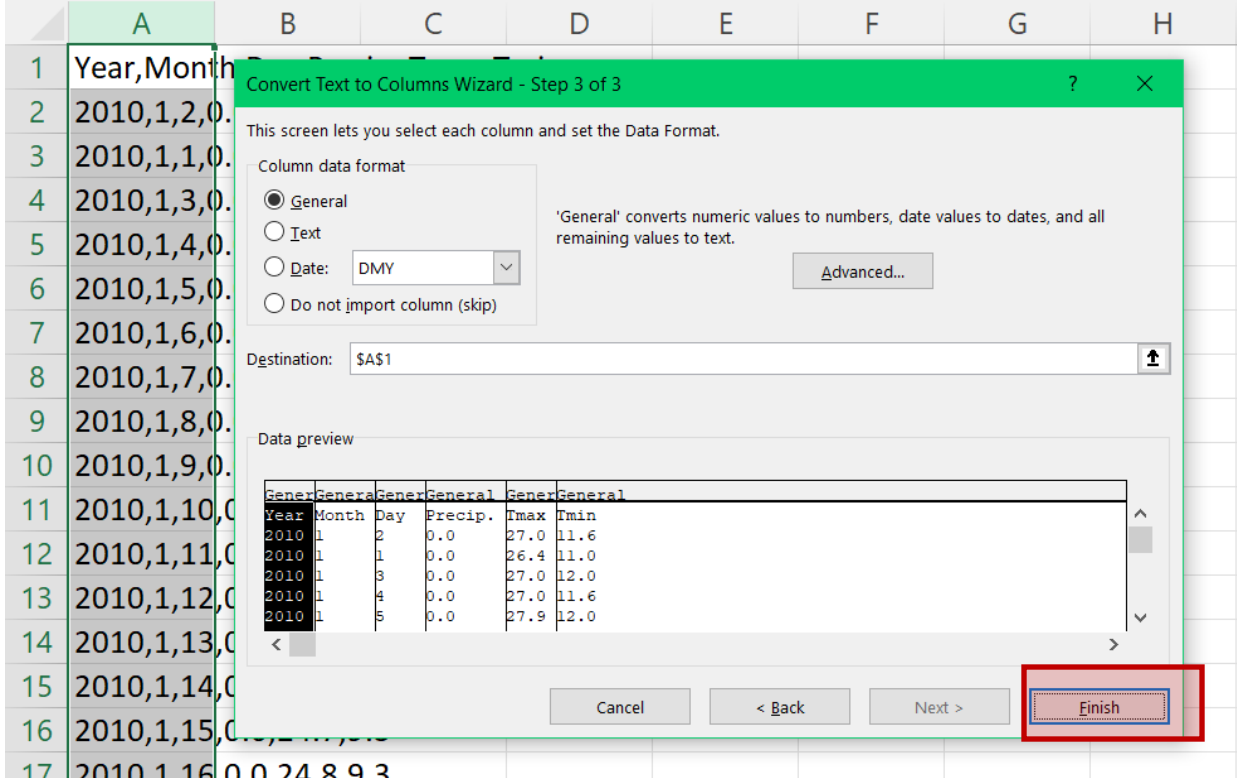


Preparing the CSV climatic files

Step 1: use the **comma** delimiter and click **next**



Step 2: click on **Finish**



Preparing the CSV climatic files

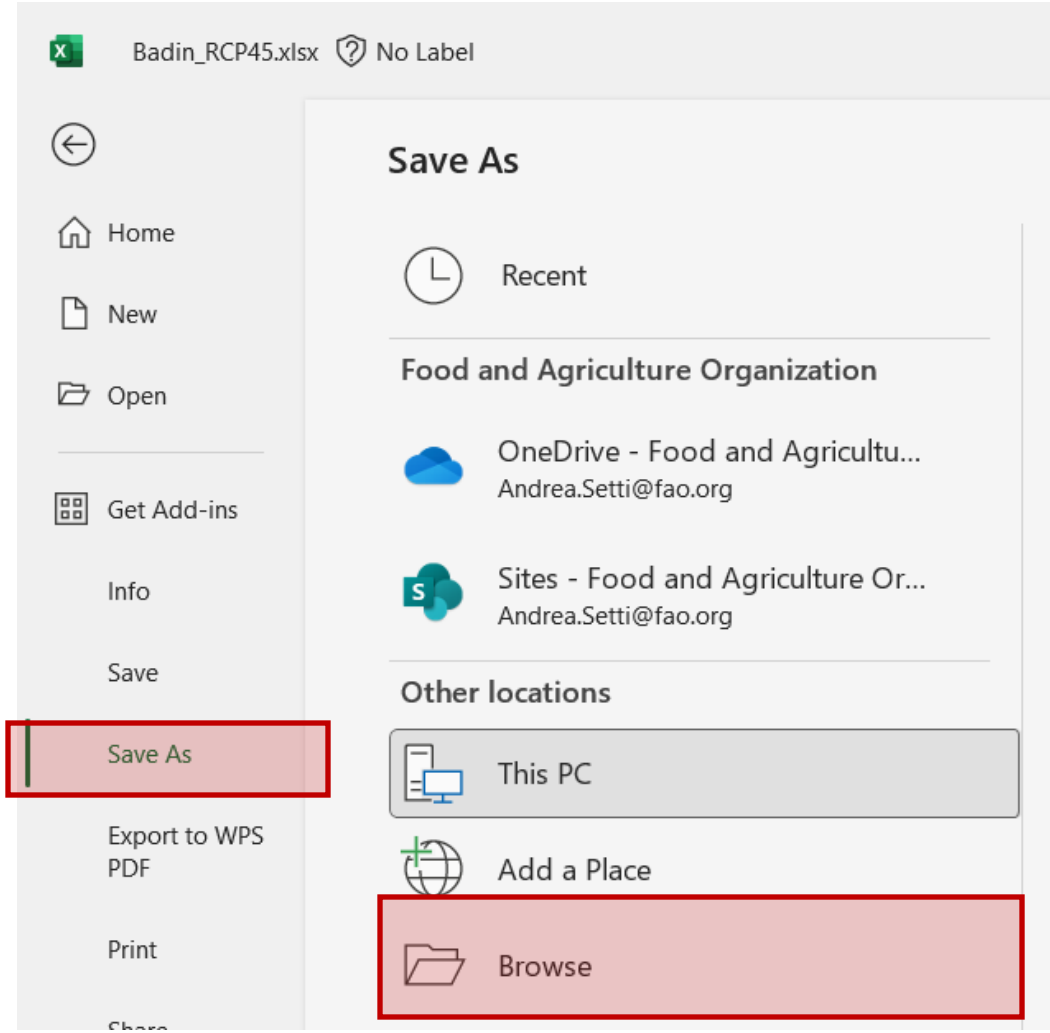
	A	B	C	D	E	F	G	H	I
1	Year	Month	Day	Precip.	Tmax	Tmin			
2	2010	1	2	0.0	27.0	11.6			
3	2010	1	1	0.0	26.4	11.0			
4	2010	1	3	0.0	27.0	12.0			
5	2010	1	4	0.0	27.0	11.6			
6	2010	1	5	0.0	27.9	12.0			
7	2010	1	6	0.0	28.6	12.0			
8	2010	1	7	0.0	28.1	12.0			
9	2010	1	8	0.0	27.0	11.3			
10	2010	1	9	0.0	26.1	10.7			
11	2010	1	10	0.1	25.4	10.9			
12	2010	1	11	0.0	25.4	11.1			
13	2010	1	12	0.0	25.1	11.1			
14	2010	1	13	0.0	24.6	10.0			
15	2010	1	14	0.0	24.7	9.5			
16	2010	1	15	0.0	24.7	9.3			
17	2010	1	16	0.0	24.8	9.3			
18	2010	1	17	0.0	25.1	9.5			
19	2010	1	18	0.0	25.3	10.8			
20	2010	1	19	0.0	24.2	10.3			
21	2010	1	20	0.0	23.7	9.1			
22	2010	1	21	0.0	24.1	8.6			
23	2010	1	22	0.0	25.2	9.5			
24	2010	1	23	0.0	26.1	10.5			
25	2010	1	24	0.0	25.6	9.9			
26	2010	1	25	0.0	25.5	9.8			

Step 1: delete the first row and the date columns (AquaCrop struggles to read text)

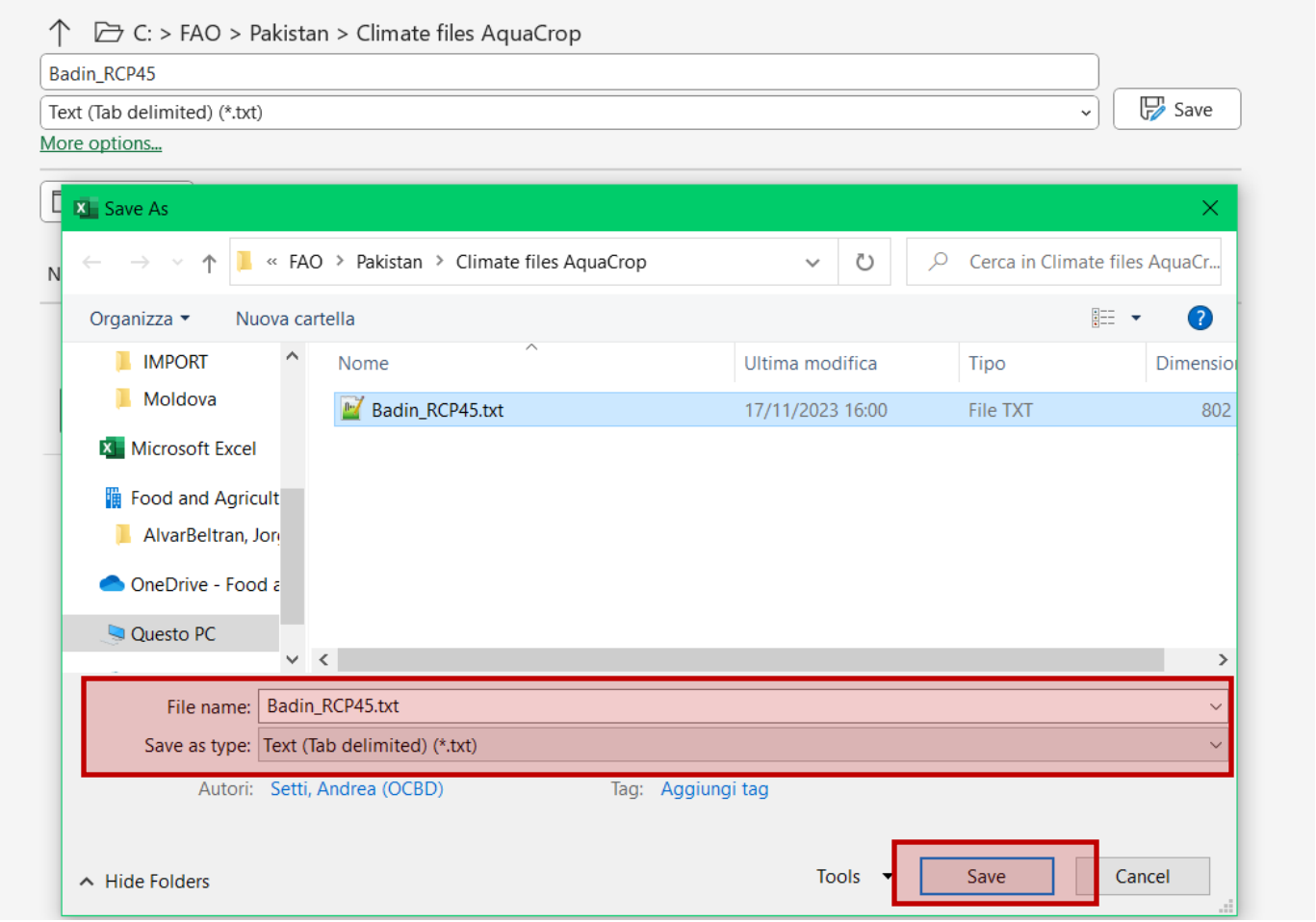
Remember that the new columns correspond to:

A) Precip.: precipitation (mm/day)
B) Tmax: maximum temperatures (ºC)
C) Tmin: minimum temperatures (ºC)

Saving the climatic files in the right format



Step 1: save the CSV file as type "**Text (Tab delimited)**"



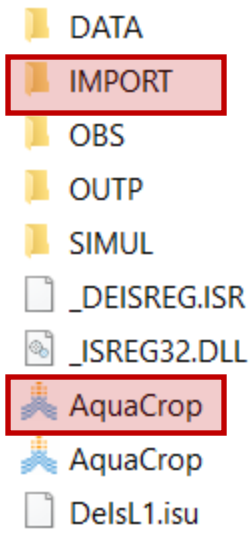
Create/import climatic files

Prec.	Tmax	Tmin
0.0	27.0	11.6
0.0	26.4	11.0
0.0	27.0	12.0
0.0	27.0	11.6
0.0	27.9	12.0
0.0	28.6	12.0
0.0	28.1	12.0
0.0	27.0	11.3
0.0	26.1	10.7
0.1	25.4	10.9
0.0	25.4	11.1
0.0	25.1	11.1
0.0	24.6	10.0
0.0	24.7	9.5
0.0	24.7	9.3
0.0	24.8	9.3
0.0	25.1	9.5
0.0	25.3	10.8
0.0	24.2	10.3
0.0	23.7	9.1
0.0	24.1	8.6
0.0	25.2	9.5
0.0	26.1	10.5
0.0	25.6	9.9
0.0	25.5	9.8
0.5	25.9	10.6
0.0	25.8	11.1

- Daily values are introduced in AquaCrop as txt. format
- The more climatic variables you have the better it is, as ETo estimations will be closer to reality…
- …and remember: AquaCrop calculates biomass production from crop transpiration.
- AquaCrop does neither read blank cells nor text
- AquaCrop has default CO_2 files from 1900 until 2100

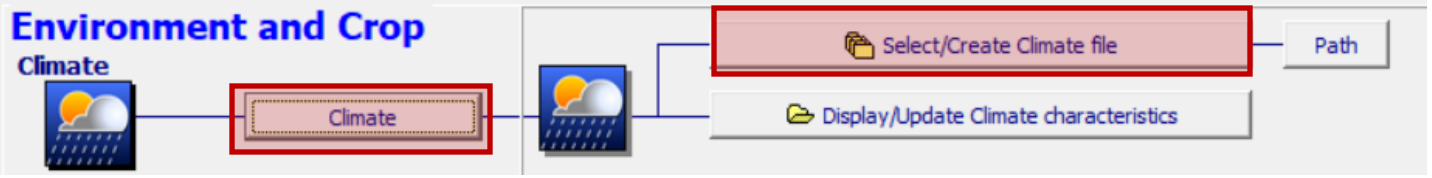
Open txt. file on AquaCrop

Step 1: copy paste the txt file into AquaCrop folder **IMPORT**

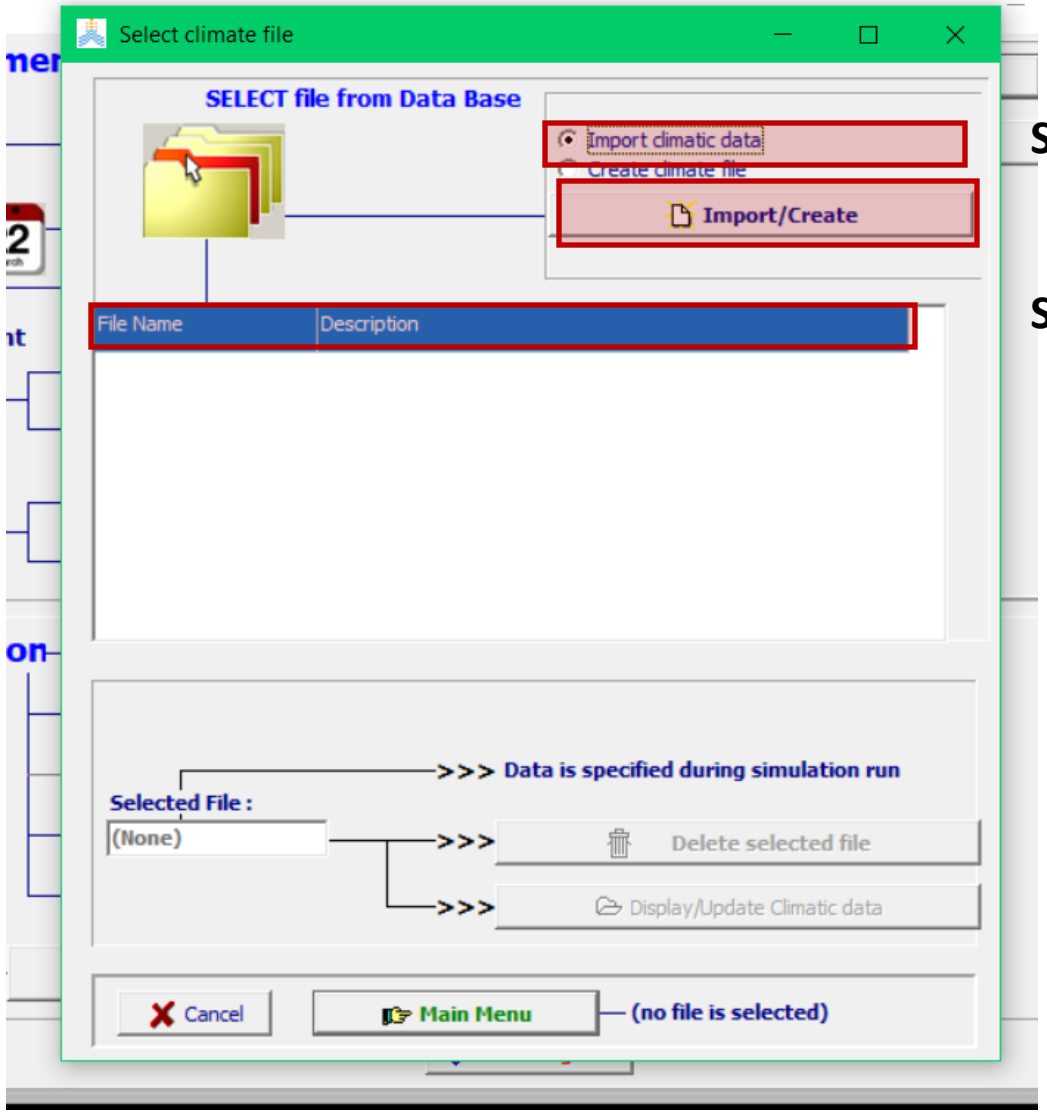


Step 2: open **AquaCrop** and click on **Start**

Step 3: open the **climate** module and click on **Select/Create Climate file**

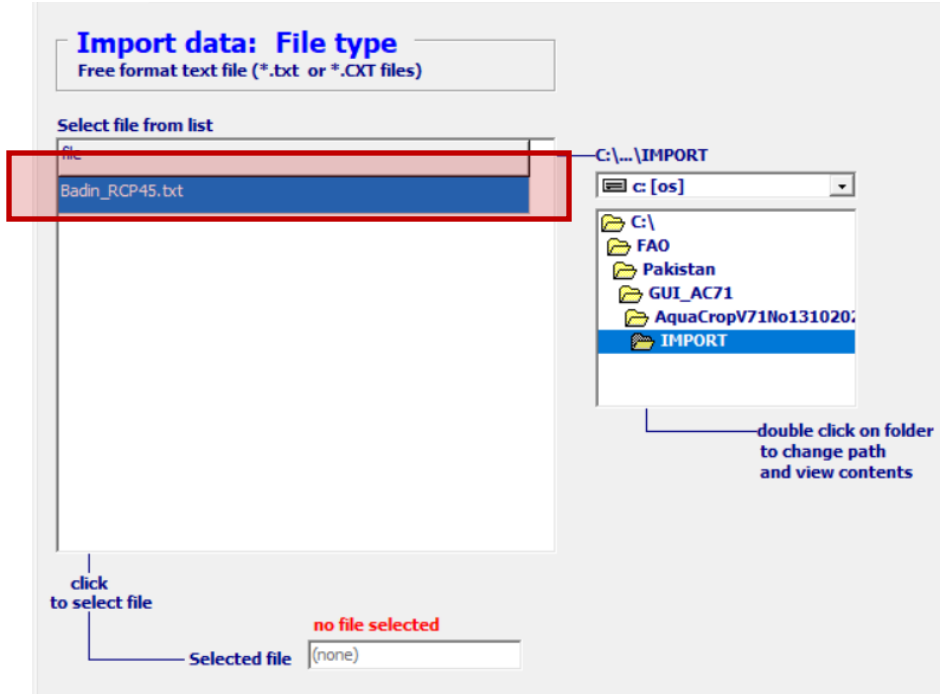


Create a climatic file

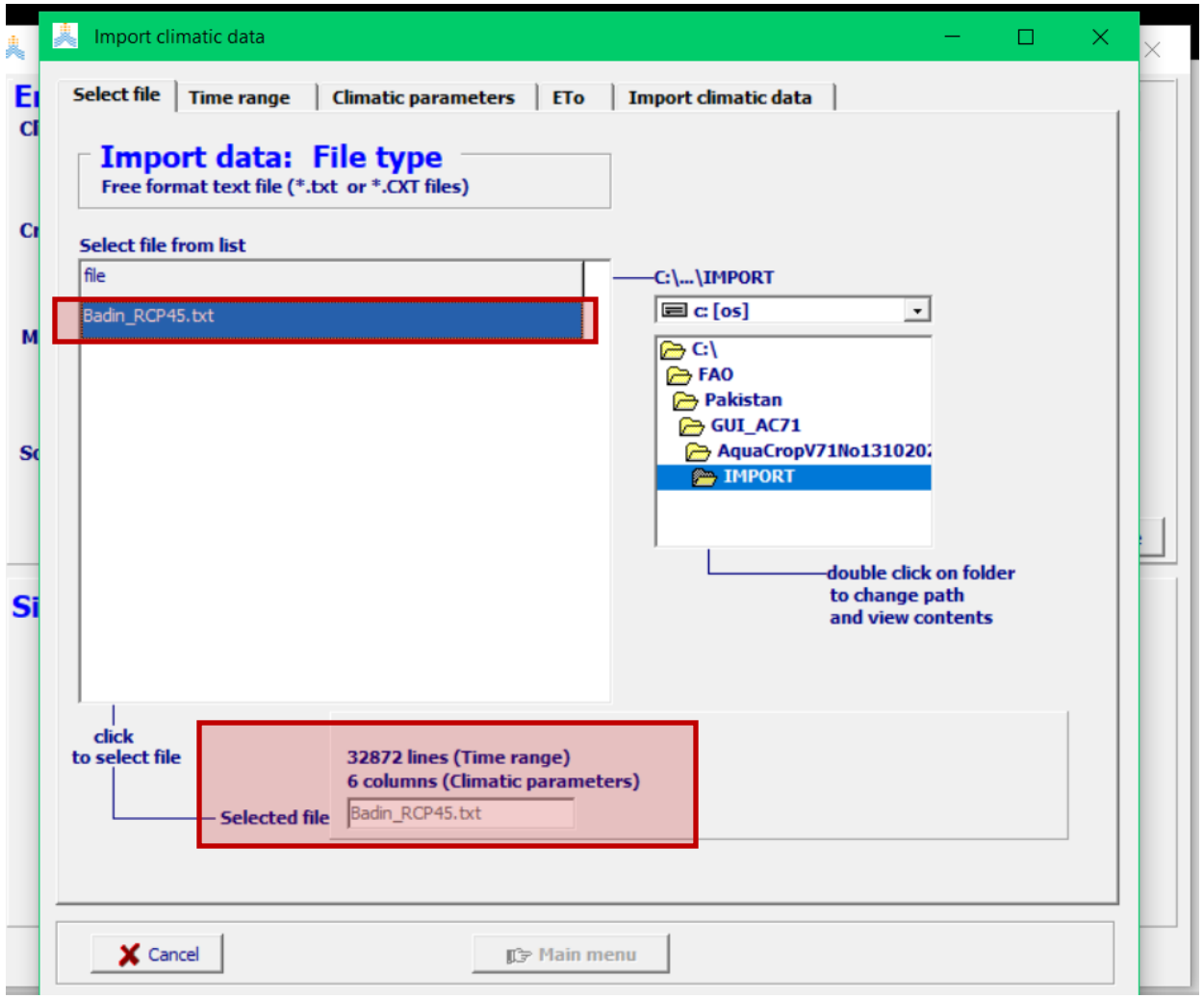


Step 1: click on **Import/Create** climatic file

Step 2: select **Badin_RCP45** and click on **Import/Create**



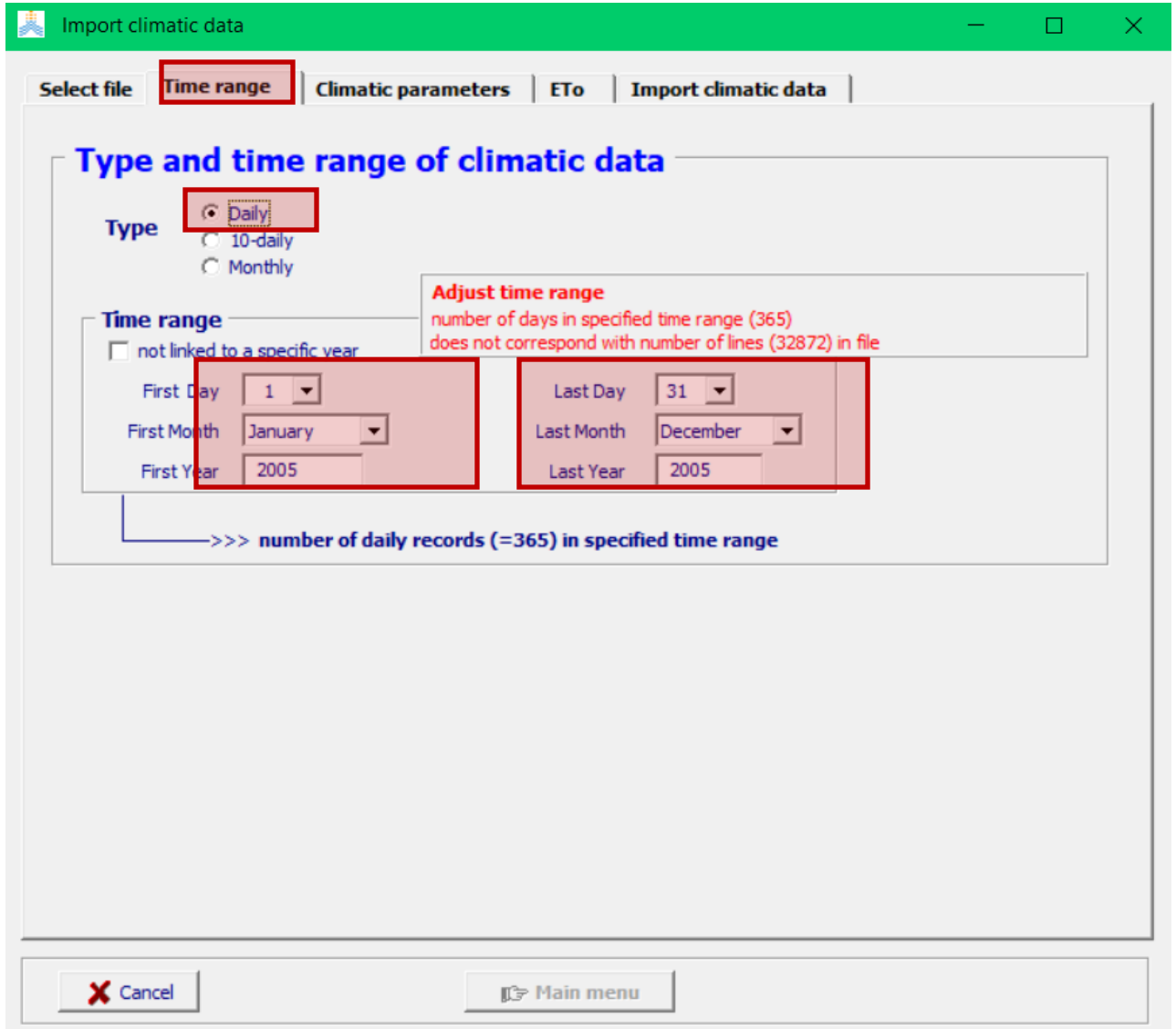
Create a climatic file



Step 1: double click on the file named **Badin_RCP45**

Double check that no errors appear in red (e.g., AquaCrop cannot read text, rows missing, blank cells etc.)

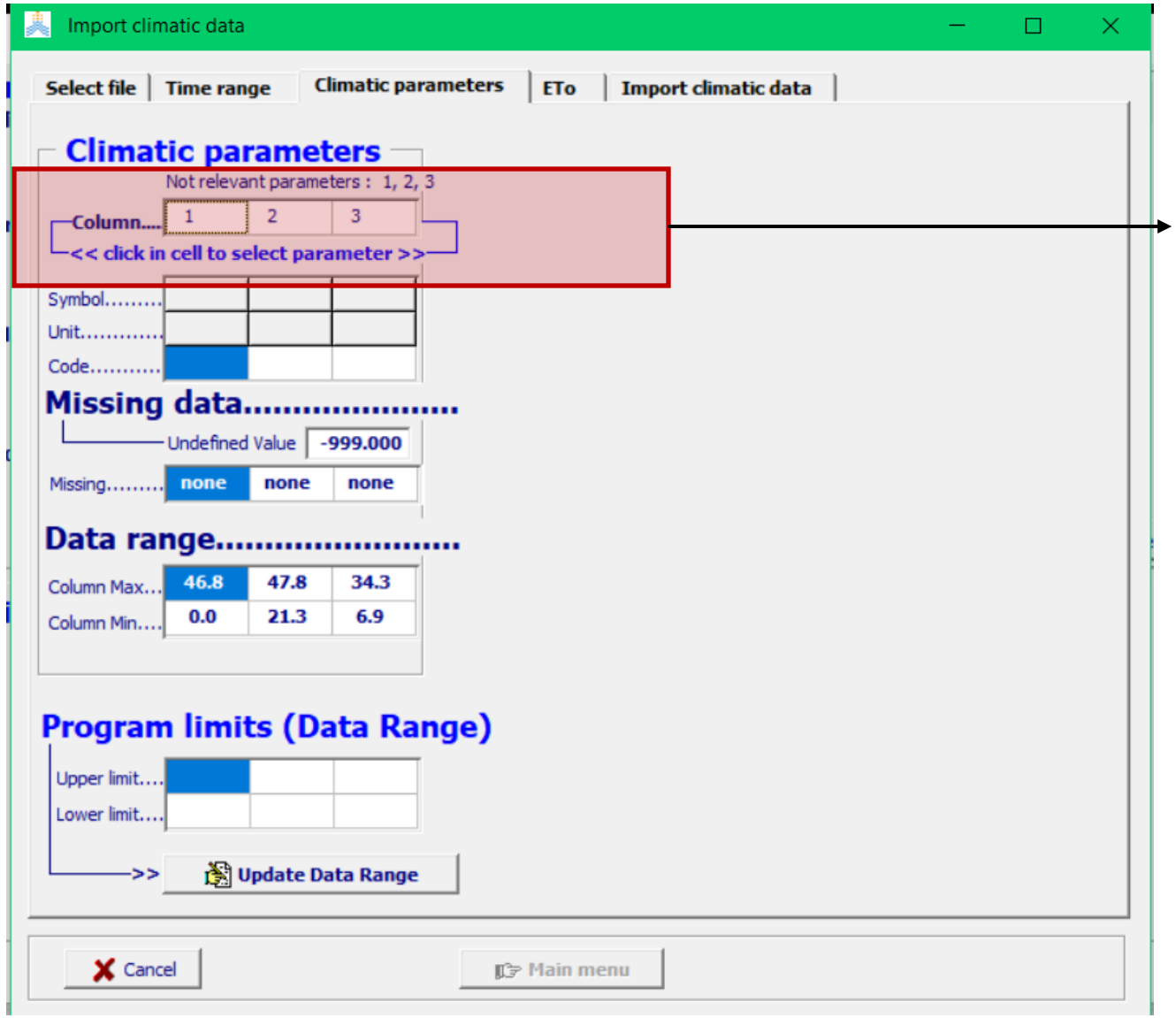
Create a climatic file



Step 1: click on **Time range** and specify the frequency of climate observations/projections. In our case **Daily**.

Step 2: select the starting **(01/01/2010)** and end period **(31/12/2099)** of daily climatic information

Create a climatic file

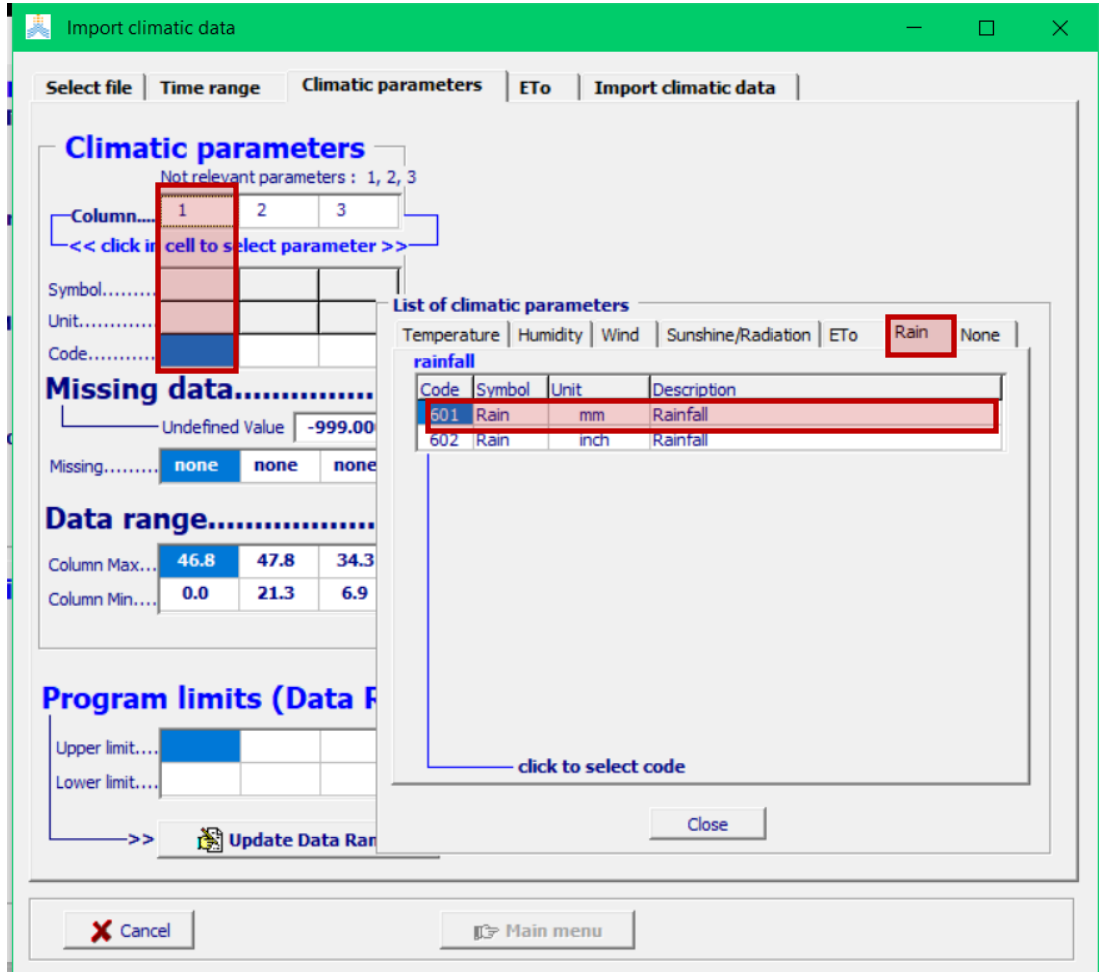


Remember that the new columns correspond to:

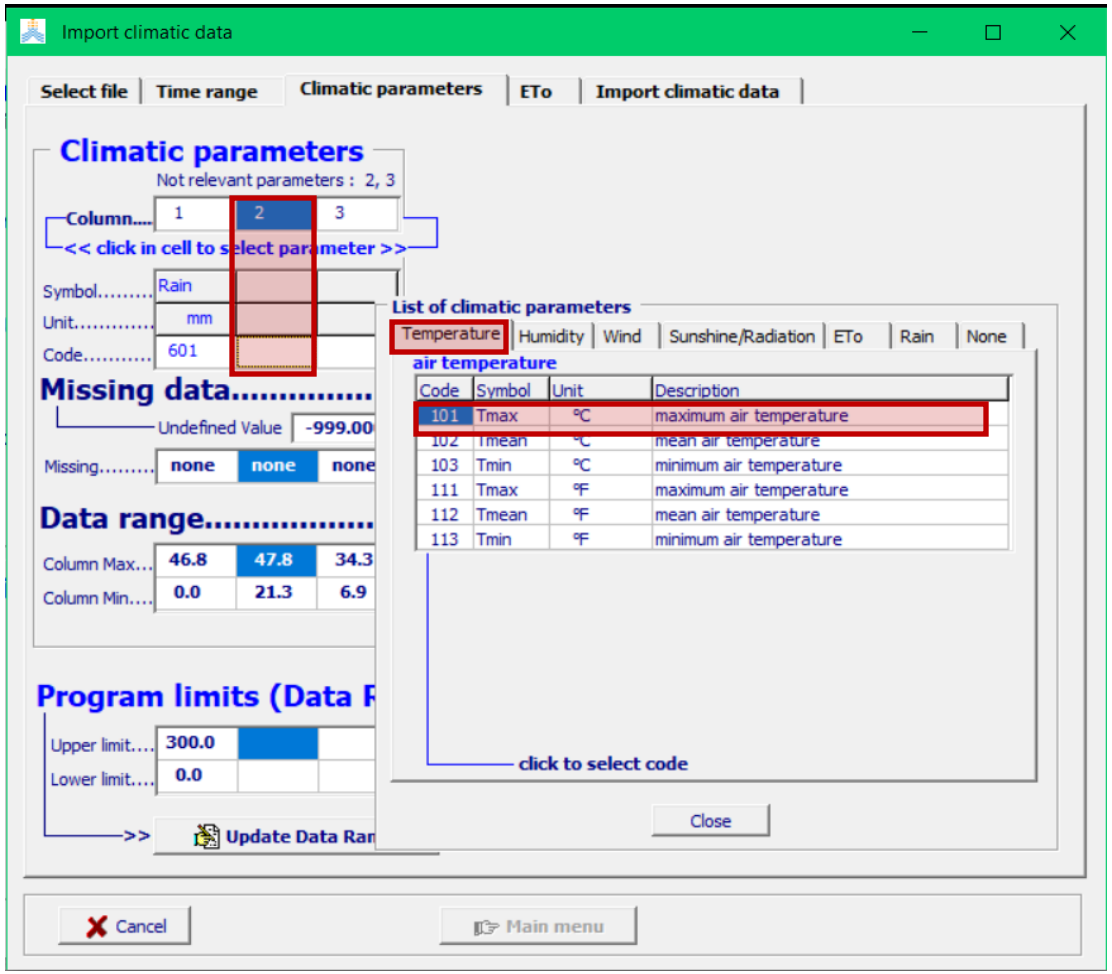
A) pr: precipitation (mm/day)
B) Tmax: maximum temperatures (ºC)
C) Tmin: minimum temperatures (ºC)

Create a climatic file (rain & Tmax)

Step 1: click on the first column (**blank cell**), then select the **rain tab** and **rain (mm).**

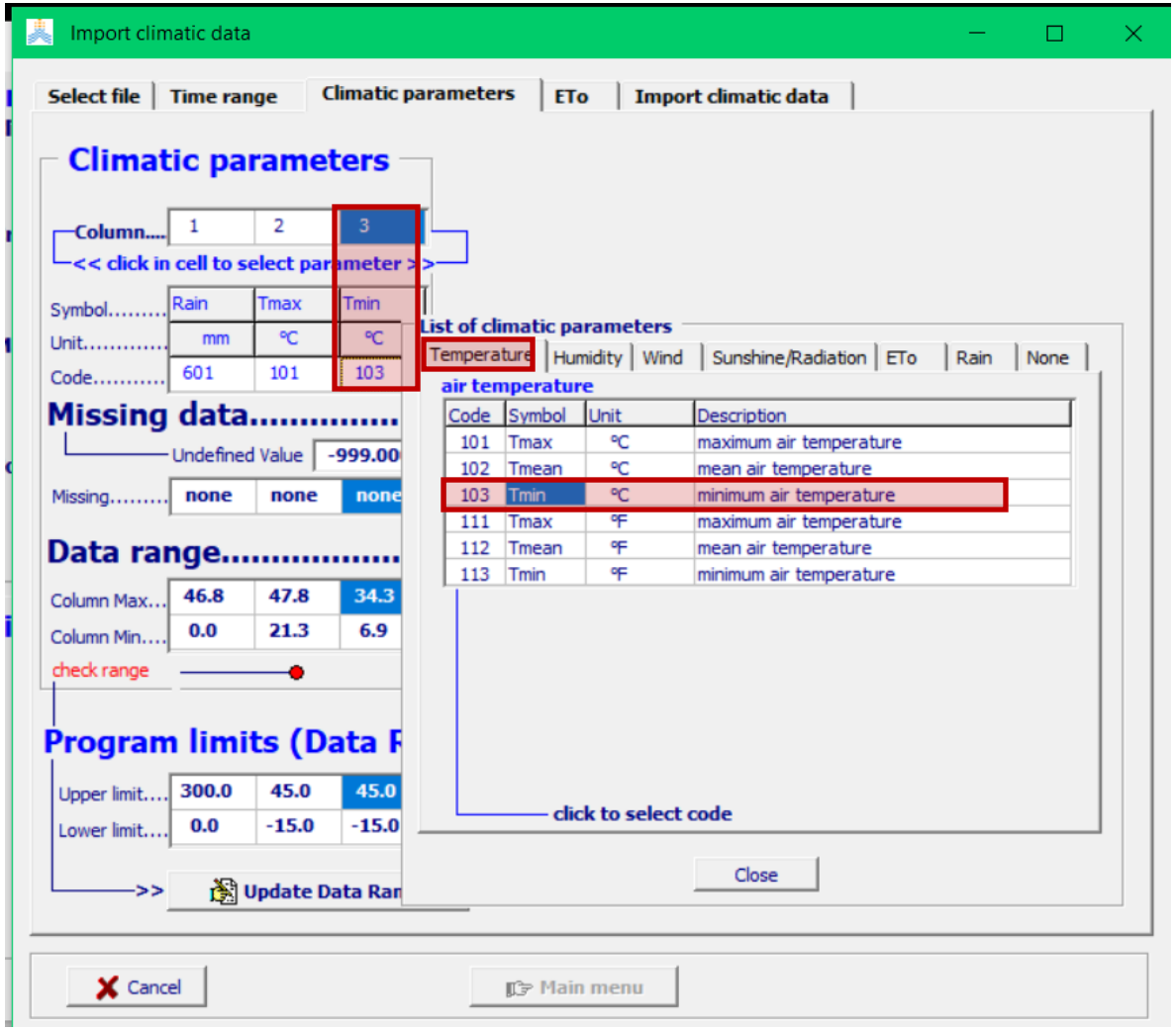


Step 2: click on the second column (**blank cell**), then select the **temperature tab** and **Tmax (ºC).**



Create a climatic file (Tmin & RH)

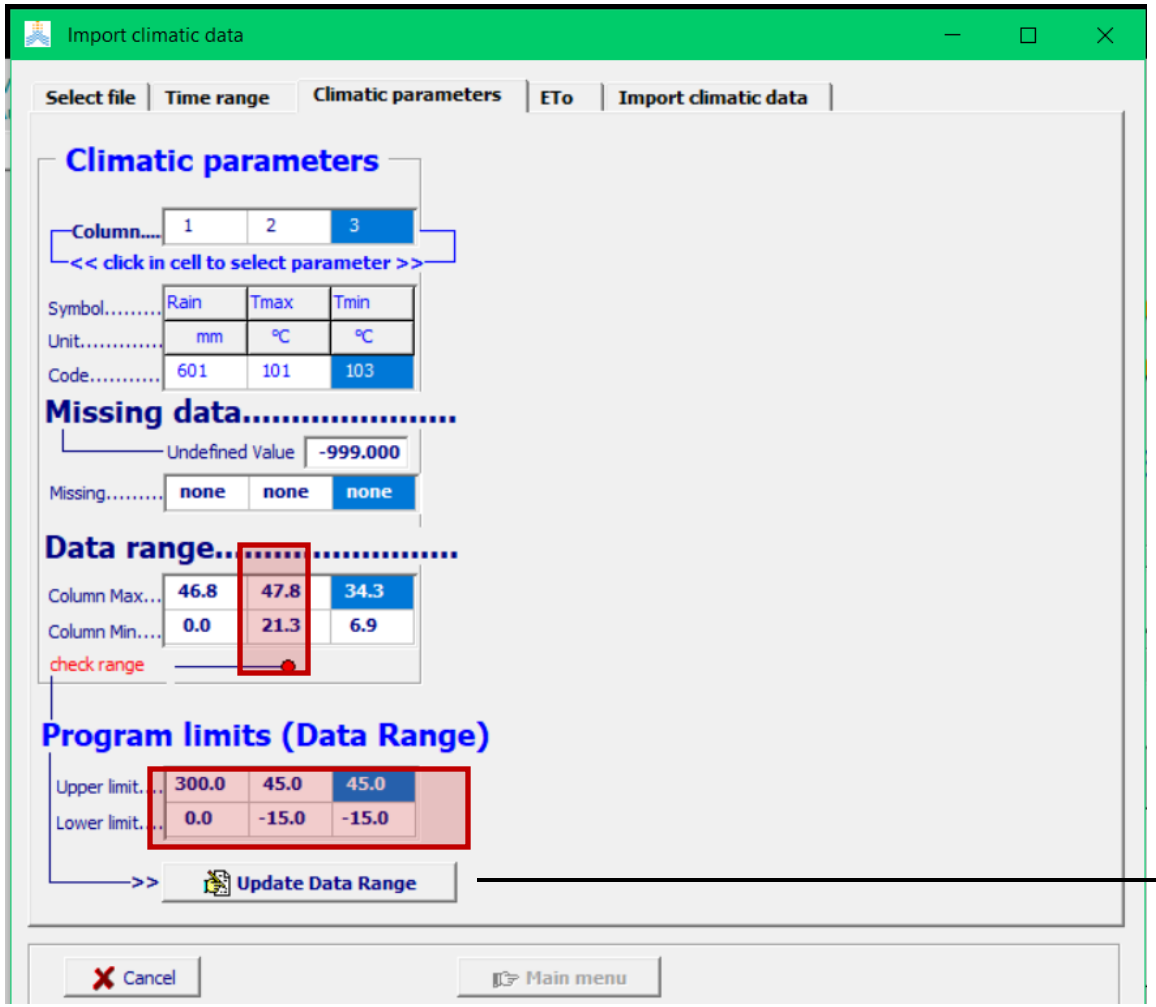
Step 3: click on the third column (**blank cell**), then select the **temperature tab** and **Tmin (ºC)**.

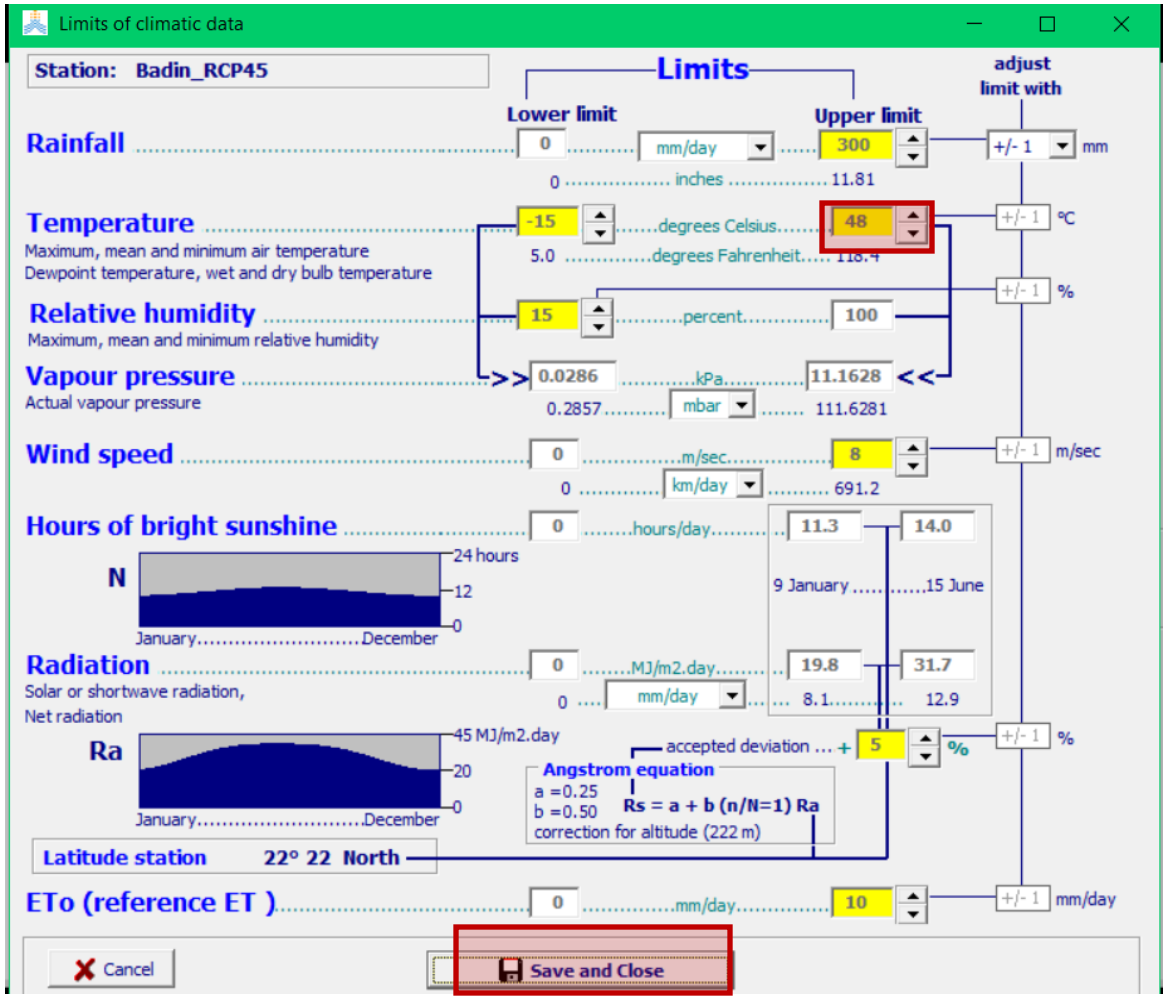


Update data ranges exceeding limits

Now you can visualize all the imported parameters as well as the data ranges for each of these variables

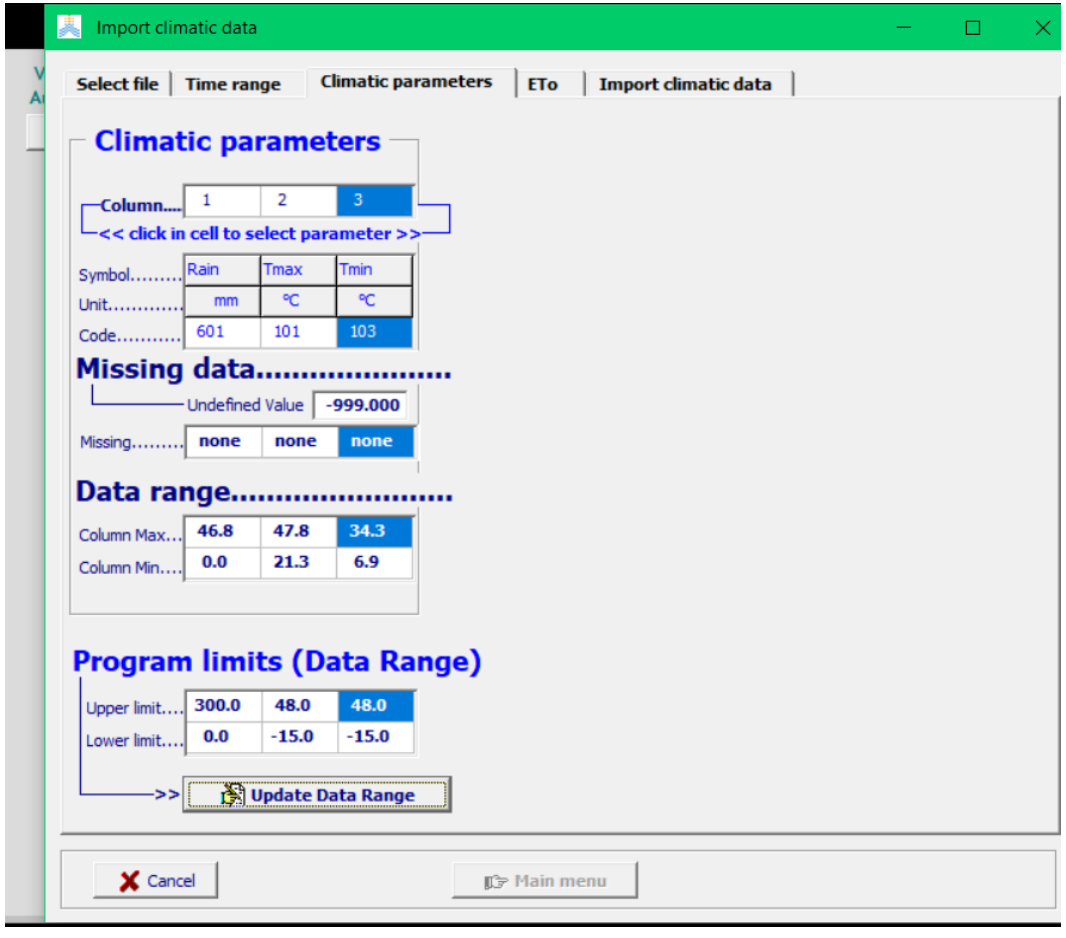
Steps 1-2: change the upper limit of **Tmax** to **48ºC** Finally, click on **save and close**



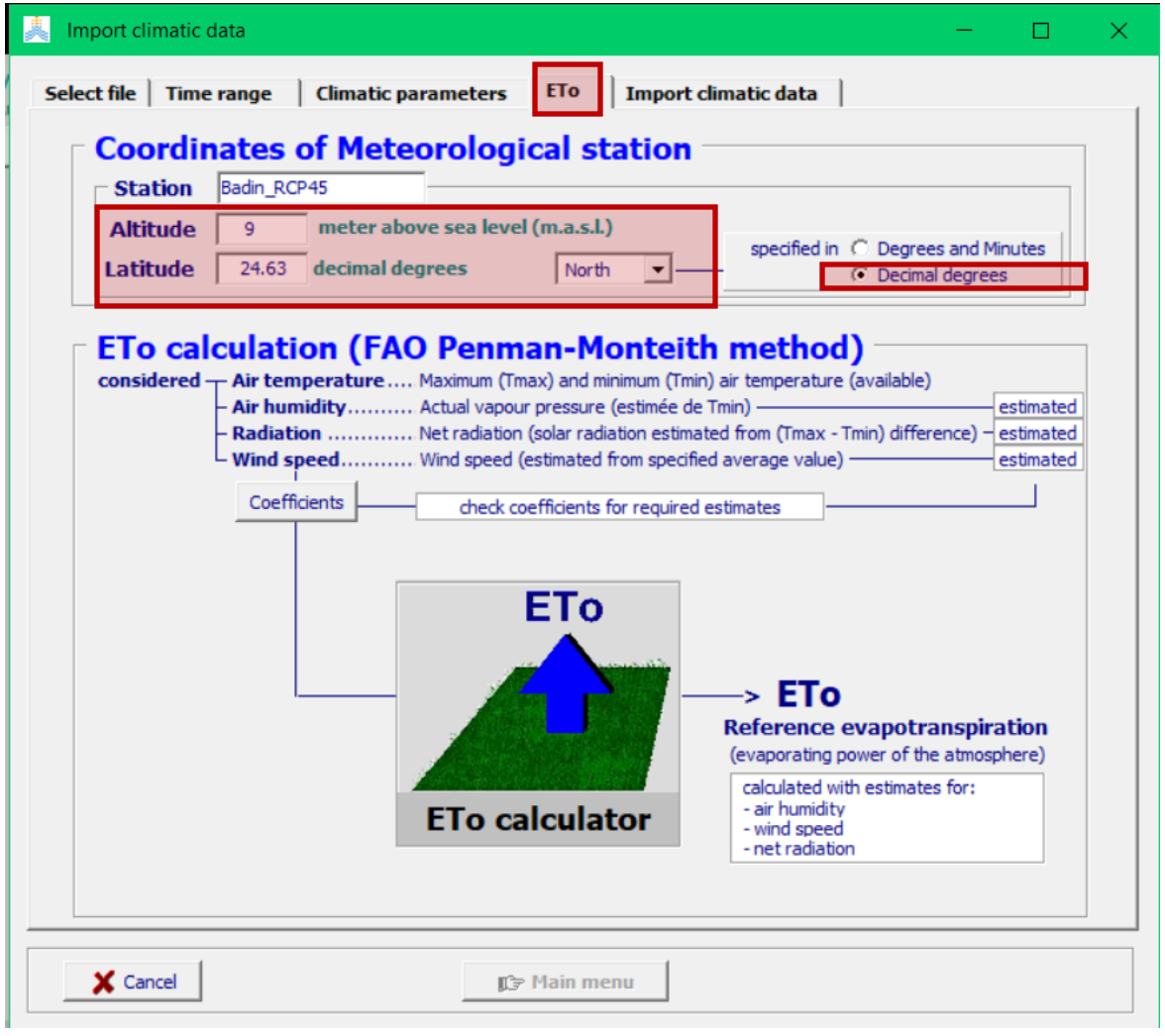


Introduce coordinates

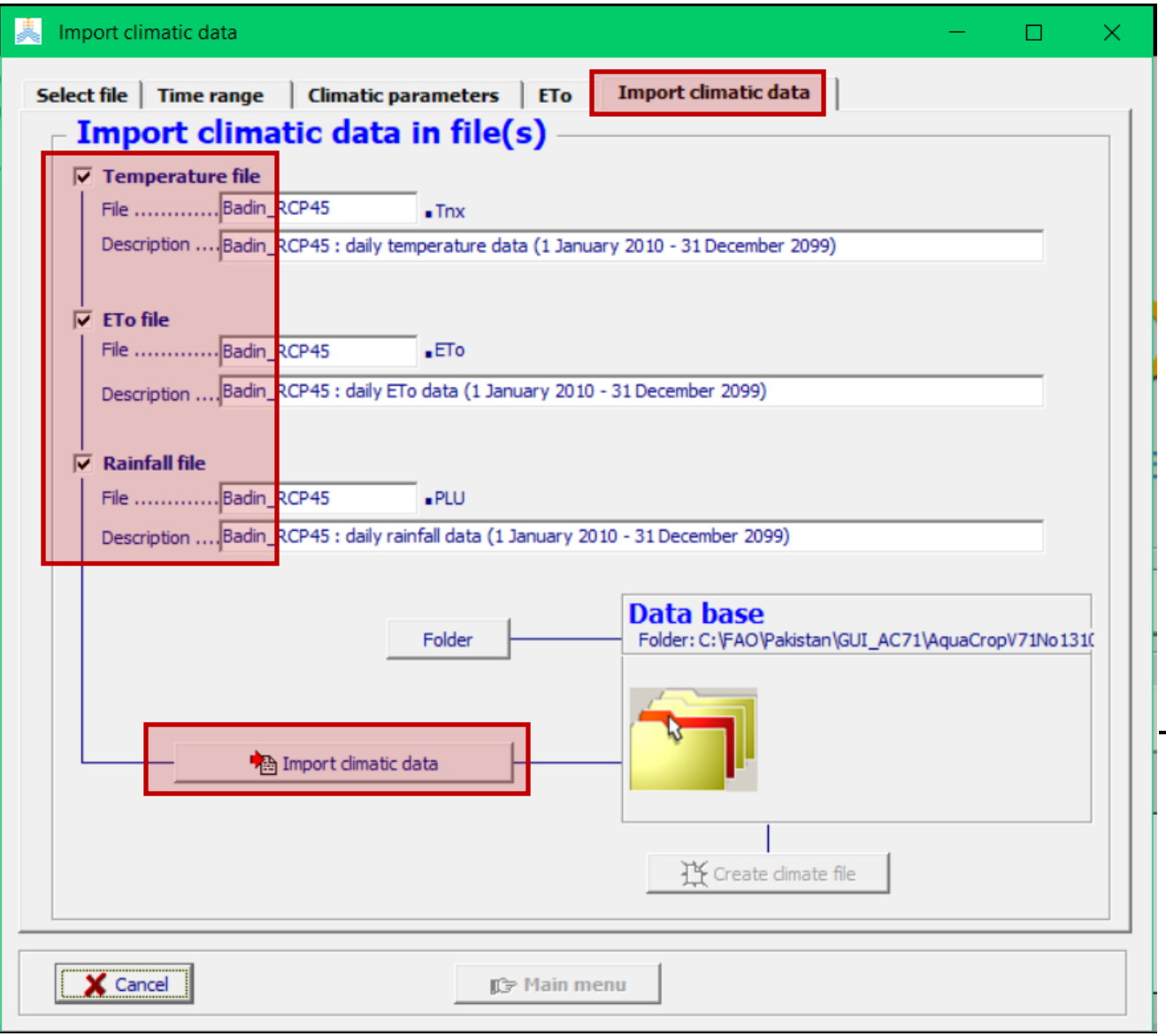
As you can see, all the imported parameters have been succesfully introduced!



Step 1: since we are preparing the climatic file for **Badin**, change the **altitude** to 9 masl and the **latitude** to 24.63 decimal degrees.

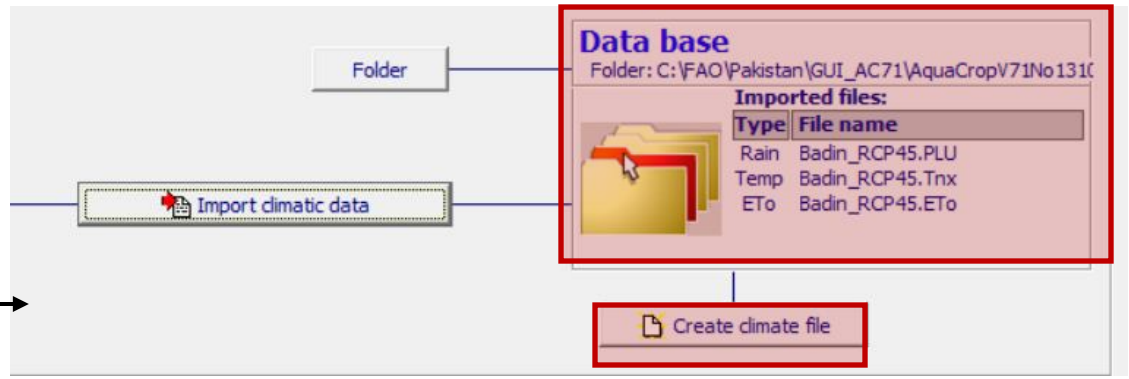


Import climatic data into AquaCrop



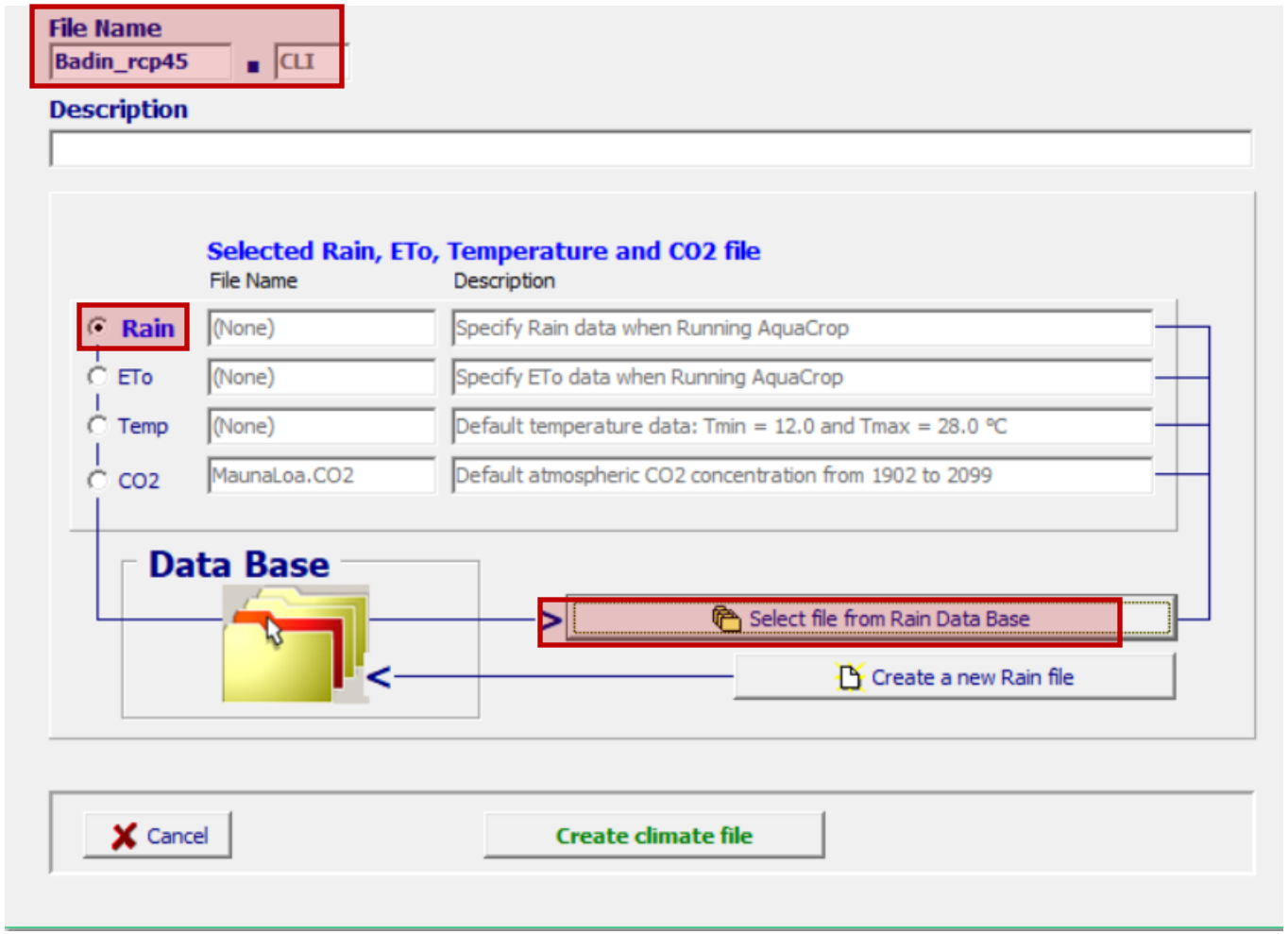
Step 1: click on **Import climatic data** and double check that the temperature, ETo and Rainfall file are selected/ticked

Step 2: click on **import climatic data** and then on **create climate file**.

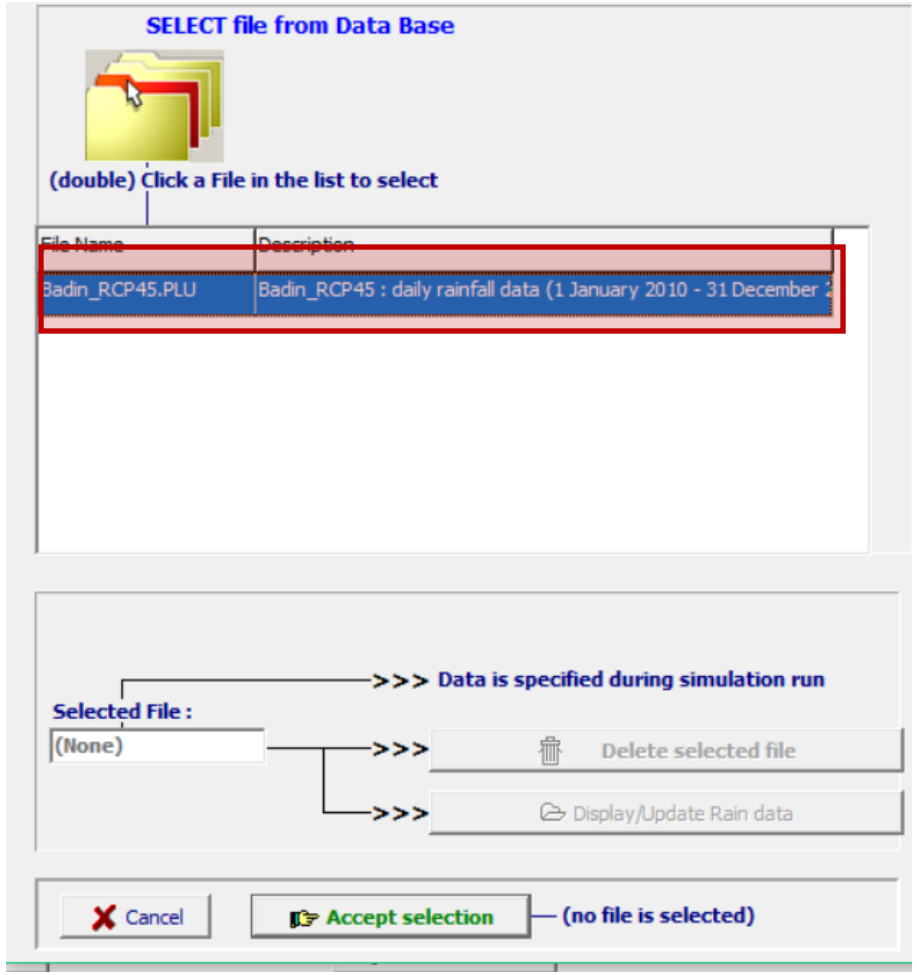


Select the rain file

Step 1. Name the climatic file as **Badin_rcp45**
Step 2. Click on **Rain** and **Select File from Rain Database**



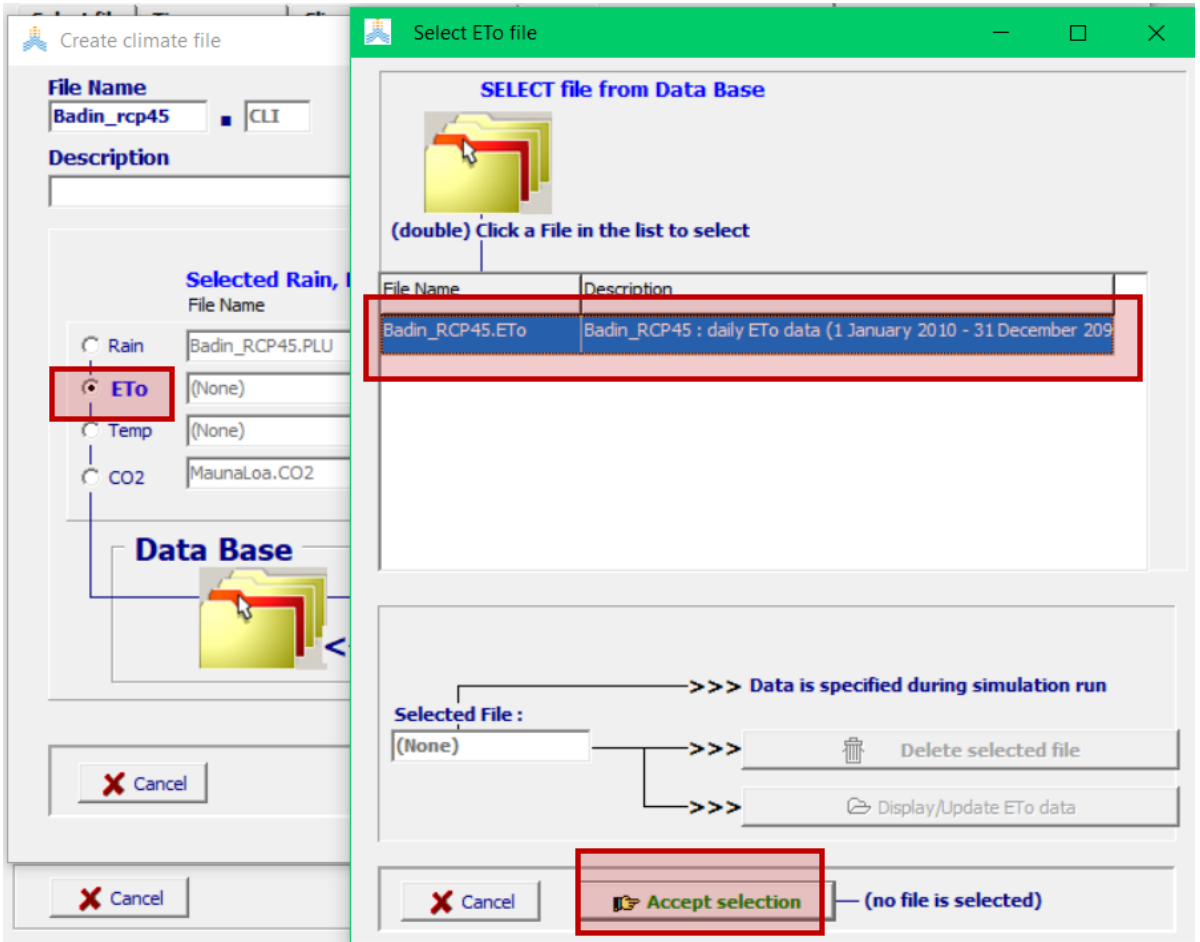
Step 3. Select the correct rain file named: **Cahul_MOHC-HadGEM2-ES_rcp26: daily rainfall**



Select the ETo and Temp files

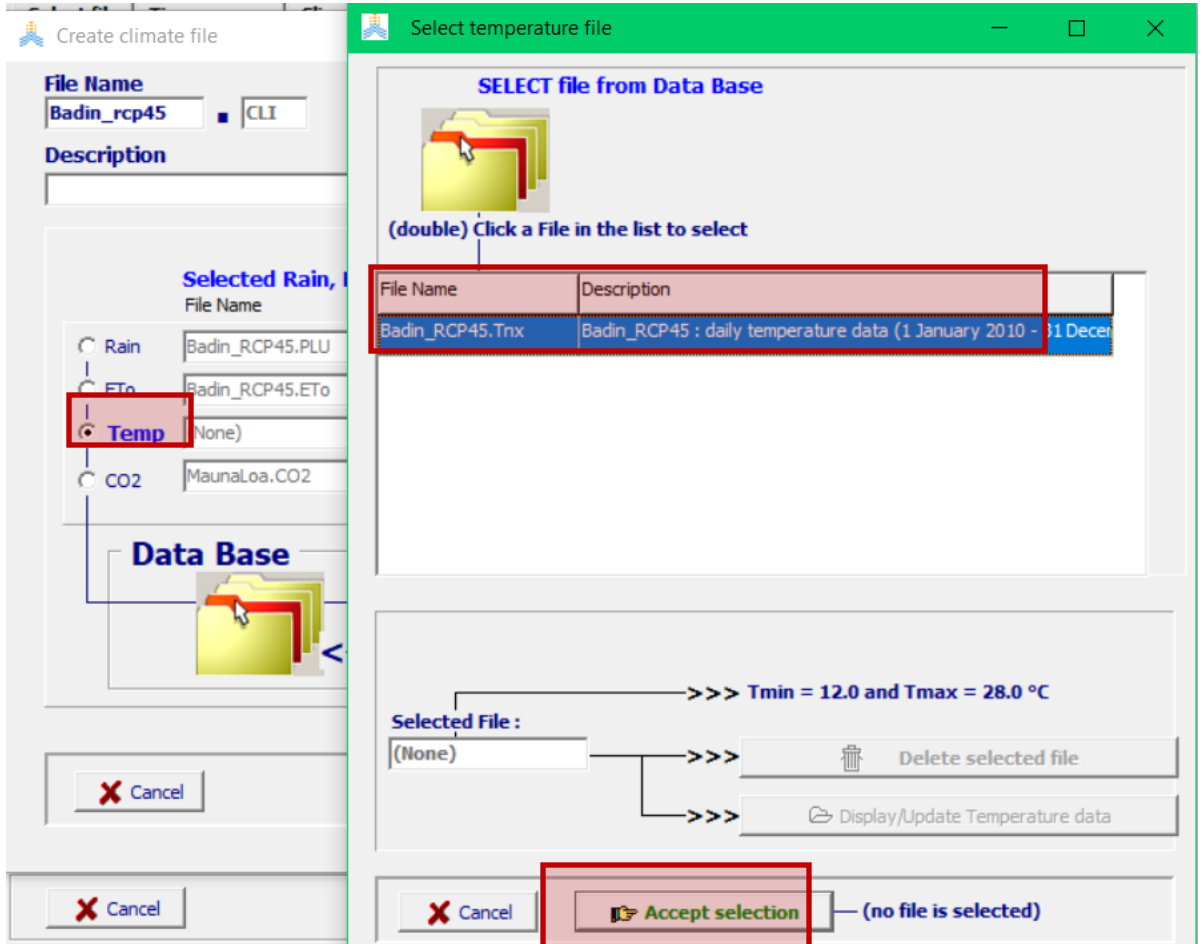
Step 1. Click on **Eto**, select file from **ETo Database** and **accept selection**

Remember that the file is named: ***Badin_RCP45: daily ETo data***



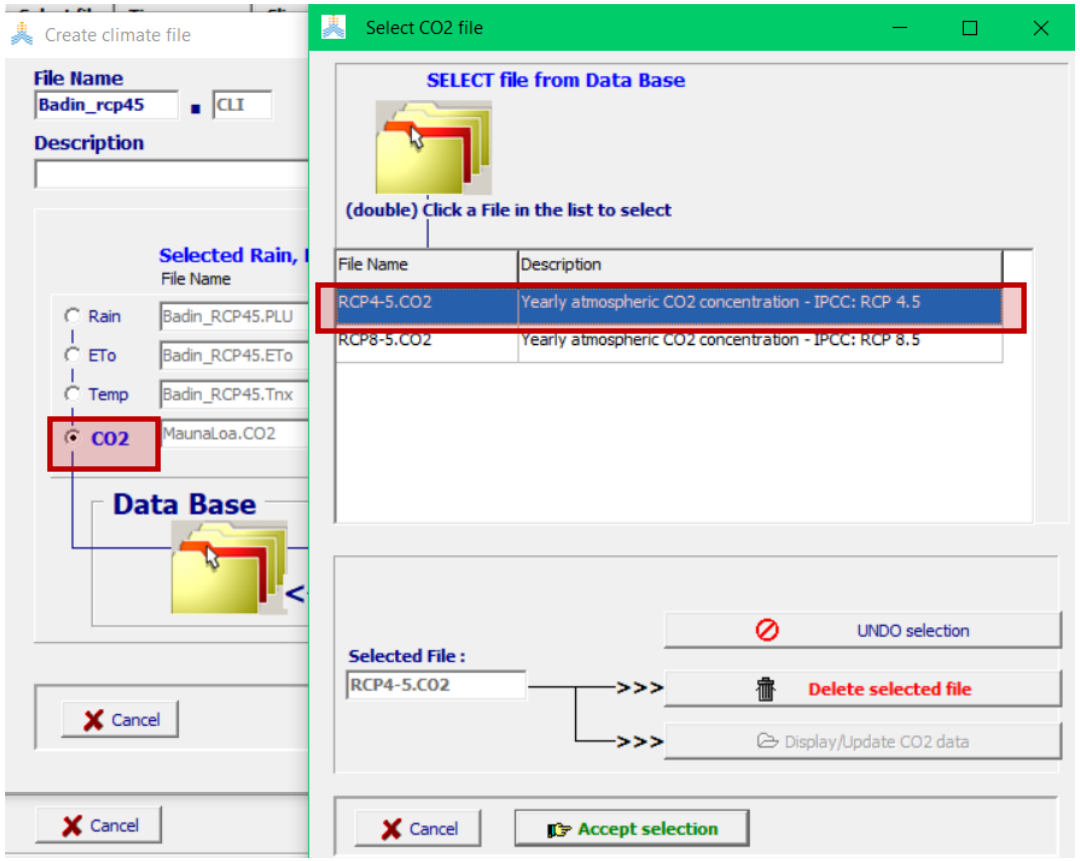
Step 2. Click on **Temp**, select file from **Temp Database** and **accept selection**

Remember that the file is named: ***Badin_RCP45: daily temperature data***

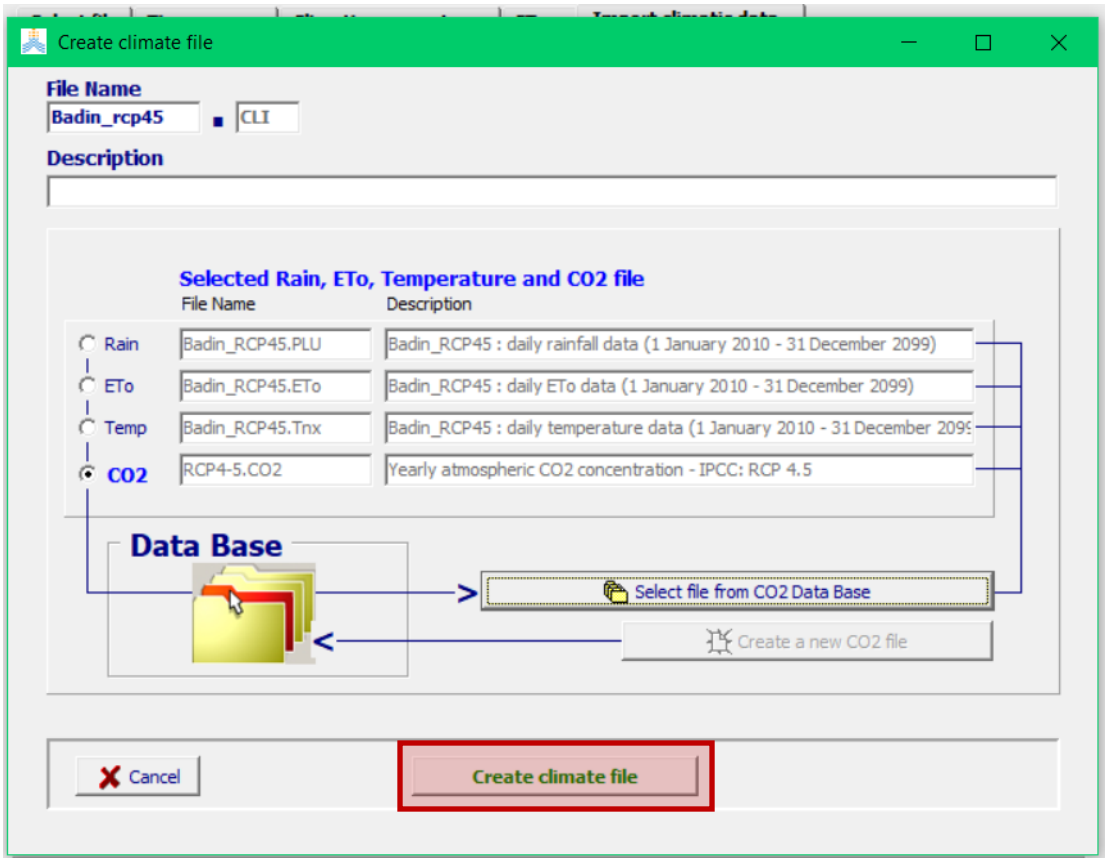


Select the CO_2 file and create climate file

Step 1. Click on **CO_2** and select **RCP 4.5-CO_2** from database



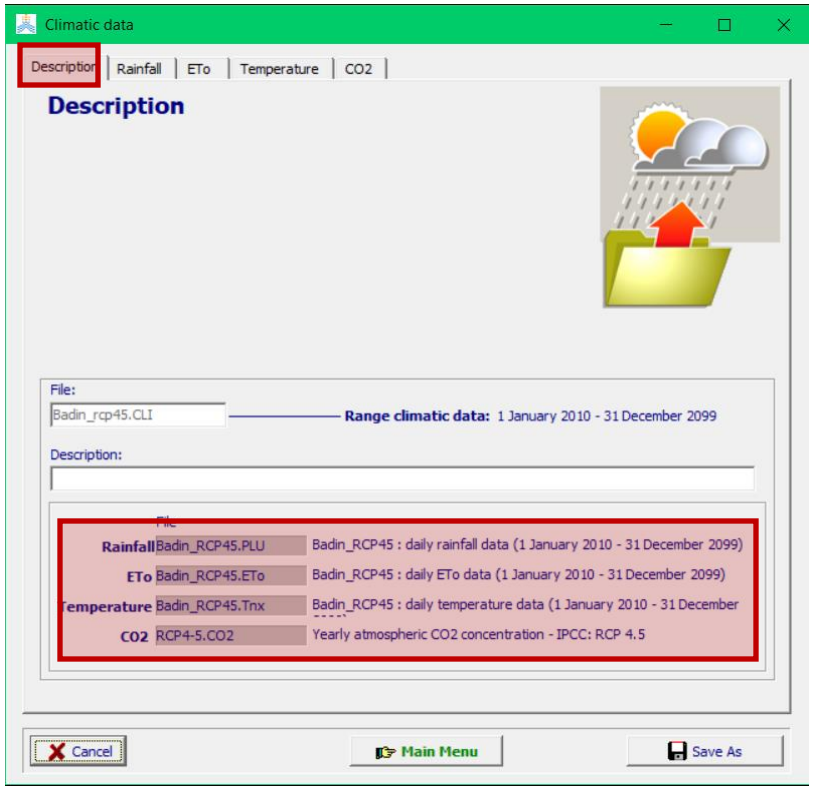
Step 2. Click on **Create climate file**



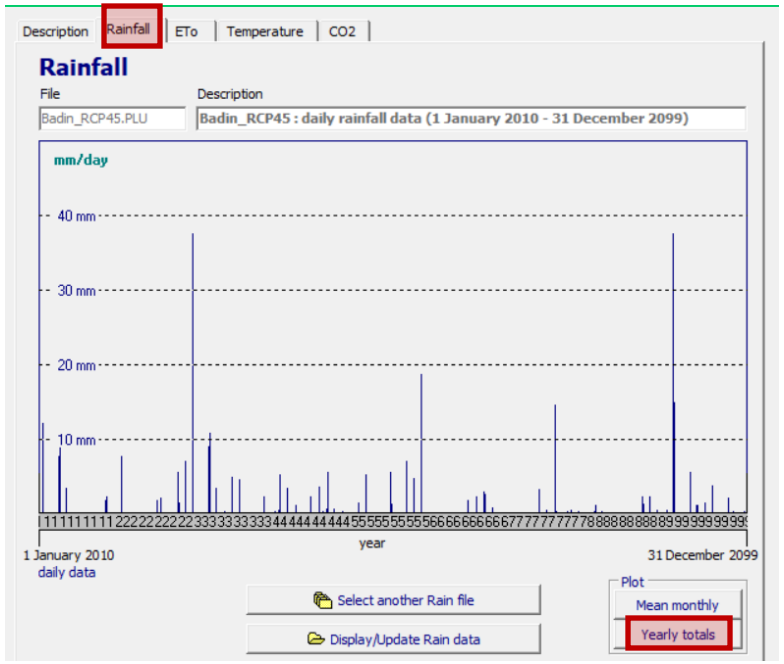
Visualize the climatic results for Badin

Step 1. Click on the **climate module** and **select Badin_rcp45**

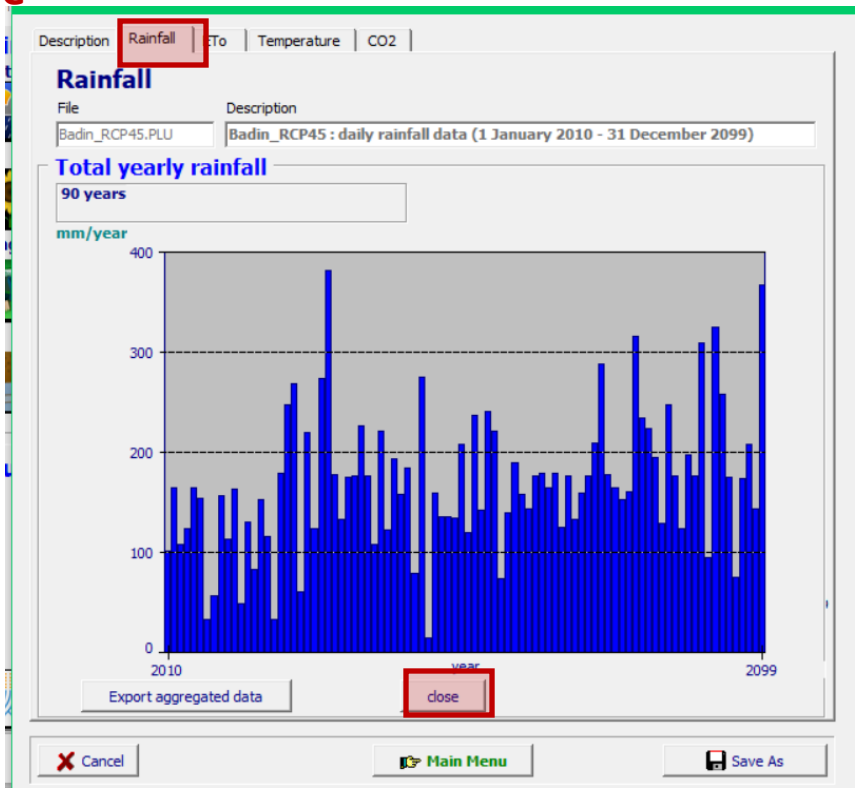
Step 1. In **description** verify that the climate files are the right ones



Step 2. Click on the **rainfall tab** and on **yearly totals**

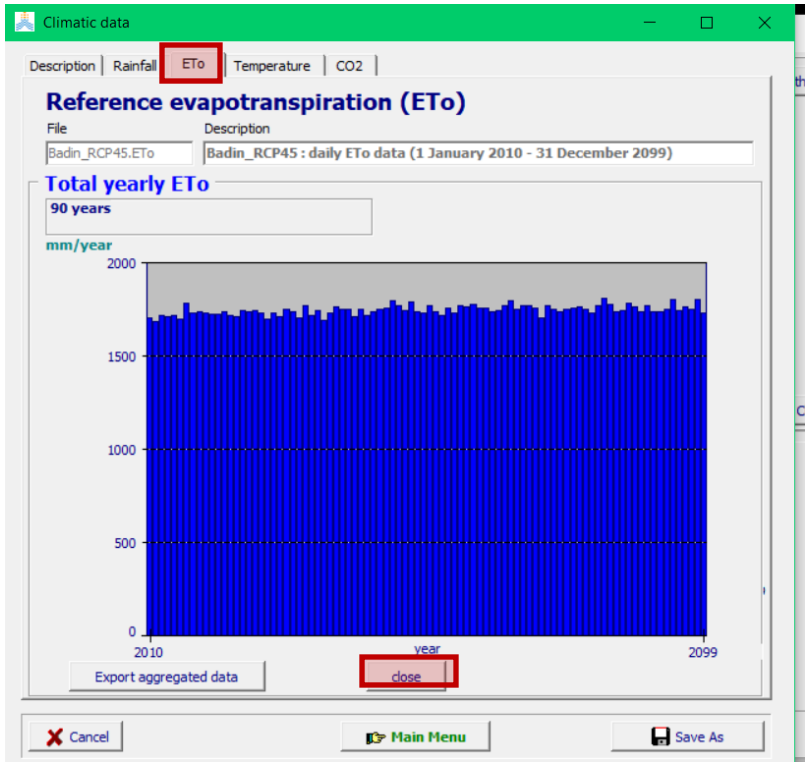


Step 3. Visualize the rainfall data → then **close**

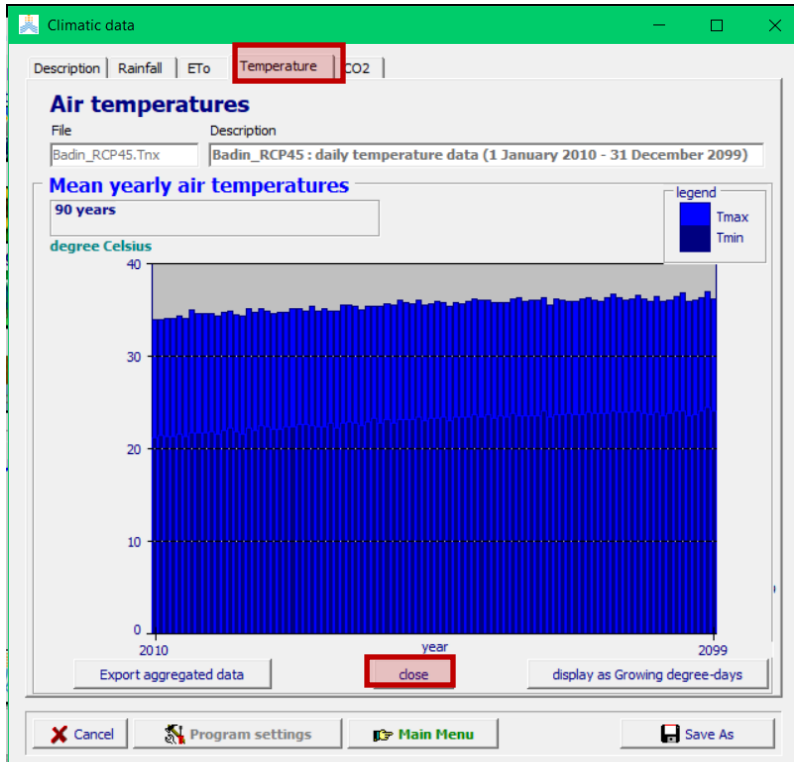


Visualize the climatic results for Cahul

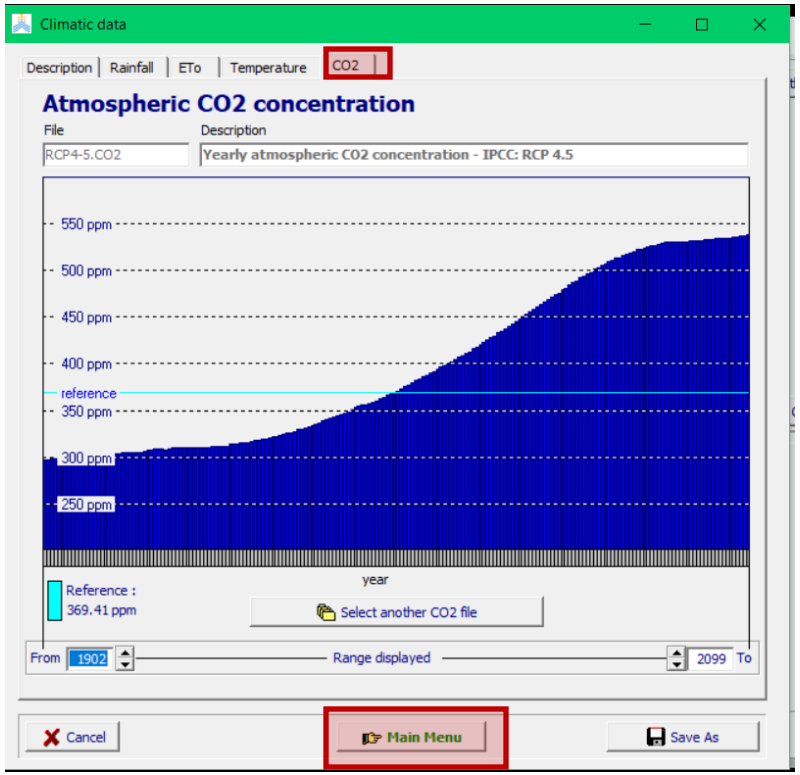
Step 4. Click on the **ETo** tab and visualize the ETo data → then **close**



Step 5. Click on the **Temperature** tab and visualize the temperature data → then **close**



Step 5. Click on the **CO_2** tab and visualize the CO_2 data for RCP 2.6 → then click on **Main Menu**



Take away messages

- Creating climatic files in AquaCrop it is a long, but straightforward process.
- If the user wants to produce daily simulations, daily weather values are required.
- To compute ETo, the user needs at least three input climatic parameters (precipitation, Tmax and Tmin).
- The maximum number of climatic parameters is six.
- AquaCrop automatically computes missing parameters (e.g., wind speed or relative humidity) necessary for running the Penman Monteith equation.
- It is important to use the right naming when saving the climatic files.
- AquaCrop automatically produces visuals for precipitation, ETo, temperature and CO_2.

Thank you!

Contact details:
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riccardo.soldan@fao.org